CPU SCHEDULING SIMULATOR

Operating Systems Tiny Project

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ABSTRACT

- Menu-driven C simulator for CPU scheduling
- Implements FCFS, SJF, Round Robin, Priority and Multi-Level Queue
- Accepts process inputs and displays scheduling results/execution order

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INTRODUCTION

- CPU scheduling selects processes for CPU execution
- Scheduling algorithms affect process order and waiting
- Project turns scheduling concepts into an interactive C program

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PROBLEM STATEMENT

- Separate manual calculations can be repetitive
- Students need a simple way to observe scheduling behavior
- Solution: one menu-driven simulator containing multiple algorithms

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OBJECTIVES

- Implement five CPU scheduling techniques
- Provide a simple menu interface
- Calculate waiting time and turnaround time where implemented
- Display process execution order
- Support practical OS learning

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METHODOLOGY

1. Display main menu
2. Select scheduling algorithm
3. Enter process data
4. Apply selected algorithm
5. Calculate/display results
6. Return to menu or exit

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CODING / IMPLEMENTATION

- Language: C
- Functions: fcfs(), sjf(), roundRobin(), priorityScheduling(), multiLevelQueue()
- Arrays store burst time, process IDs, priorities, queues and remaining time
- switch statement connects menu choices to functions

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HARDWARE & SOFTWARE REQUIREMENTS

- Hardware: Laptop/Desktop, keyboard, monitor, 4 GB RAM recommended
- Software: GCC/C compiler
- IDE: VS Code / Code::Blocks / Dev-C++
- OS: Windows or Linux

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PROJECT OUTPUT

- Add screenshots here:
  - Main menu
  - FCFS result
  - SJF result
  - Round Robin execution order
  - Priority result
  - Multi-Level Queue execution order

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ADVANTAGES

- Simple menu-driven operation
- Multiple algorithms in one project
- Reduces repetitive manual calculations
- Useful for OS laboratory learning
- Easy to compile and modify

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LIMITATIONS

- Educational simulator, not a kernel scheduler
- Simplified process inputs
- Multi-Level Queue demonstrates queue ordering
- Command-line interface only

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CONCLUSION

- Demonstrates practical CPU scheduling in C
- Connects OS scheduling concepts with implementation
- Provides an easy way to observe scheduling results
- Thank You / Questions?

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